

COMP1047 AY2526: Final Exam Review Guideline

- **NO** calculator is allowed.
- We will provide you the MIPS Ref Card appended at the end of the exam paper.
- Dictionary is permitted as per usual exam rules.
- The final exam covers the two (2) aspects that have been taught in the lecture and exercised in the labs:
 - Computer Architecture
 - Basics and MIPS programming
 - ISA and CPU Design
 - Computer Networks

Basic concepts and MIPS programming

- Week 1: Introduction & Computer Performance
 - General concepts
 - Bit, byte, word
 - CPU timing: clock period, frequency, CPI, etc.
 - von Neumann architecture, ISA, microarchitecture, etc. -> Also in Week 4.
- Weeks 2 & 3: MIPS programming 1 & 2
 - All concepts: registers, memory (address, data, addressing, endianness), immediate, etc.
 - and related instructions: arithmetic, memory, logical, branch/jump, etc.
 - Arrays and procedures, **but no recursive procedures.**
 - **Syscall will not be covered.**

ISA and CPU design

- Week 4: MIPS ISA
 - R-, I-, J-types instructions – definitions, design, and conversion. MIPS ref card will be provided for assistance.
 - Content in the handout may be included, too.
- Week 4: ALU Design
 - All concepts including
 - Basic hardware principles and functionalities
 - Hardware details will not be covered, such as 1-bit adder logical expression, multi-bit adders design.
 - ALU design principle, schematic, and control
 - Add, Sub, And, Or, Slt, Nor, Nand, etc.
- Week 5: Single-Cycle CPU design
 - All concepts including
 - Drawing the datapaths for R-type, lw, sw, beq instructions.
 - You don't need to know how the controller is designed, but need to know what are the corresponding control signals required to make the components work properly.